

shopEZ - Full-Stack E-Commerce Marketplace (MERN)

This document provides a comprehensive walkthrough of the features and architectures implemented in the **shopEZ** platform, verifying compliance with the system specifications.

1. Architecture & Core Tech Stack

- **Backend (Node.js & Express):** Located in [backend/](#). Centralized error handling, input validation, and RESTful routing structure.
 - **Database (MongoDB/Mongoose):** Models defined in [backend/models/](#) representing:
 - [User](#) (with shipping addresses, wishlist array, and roles)
 - [Product](#) (with rating aggregation, variants, and stock control)
 - [Order](#) (with pricing breakdowns, status timeline, and payment info)
 - [Review](#) (verified purchase indicator, star rating, and helpful vote tracks)
 - [Coupon](#) (active dates, type, discount values)
 - **Database Seeder:** The application features a dynamic seeder [db.js](#) which starts an **In-Memory MongoDB fallback server** via [mongodb-memory-server](#) if a local connection isn't available. It auto-seeds the system with mock customers, products, reviews, and active coupon codes.
 - **Frontend (React + Vite + Tailwind CSS):** Premium, responsive, mobile-first design using HSL tailored dark-and-light-mode compatible palettes, Outfit/Inter typography, and subtle transitions. Built using functional components and Context state providers.
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2. Core Feature Walkthrough

2.1 User Authentication & Authorization (RBAC)

- **Register / Login:** Secure registration and session creation with JWT tokens.
- **Role-Based Access Control:**
 - Enforced in the backend via JWT checking in [authMiddleware.js](#) ([protect](#), [admin](#)).
 - Guarded in React routes via [ProtectedRoute.jsx](#).
- **Profile Dashboard:** Built in [ProfileDashboard.jsx](#). Users can edit profiles, manage multiple saved shipping addresses (add, edit, remove, set default), and view account configurations.

2.2 Product Catalog & Search/Filters

- **Listing & Details:** High-fidelity product card layouts with ratings summaries, discount/sale badges, stock warnings, and specification lists.
- **Detail View:** Interactive thumbnail selector on the product detail page with related products carousel, stock status, delivery estimation checker, and double-review prevention.
- **Full-Text Filter & Search:** Custom search inputs with category dropdown select, price ranges, minimum ratings filter, availability toggle, and sort configurations (low-to-high, high-to-low, newest, ratings).
- **Autocomplete Suggestions:** Auto-filters based on live name inputs.

2.3 Cart & Wishlist Lifecycle

- **Persistent Cart:** Integrated with Context state. Handles quantities, stock checks, and tax calculations.
- **Saved Wishlist:** A custom space where users add/remove favorites. Includes a drop-in option to "Move to Cart" which transfers items and clears them from the wishlist in one tap.

2.4 Checkout & Mock Stripe Payments

- **Multi-Step Checkout:** Addresses, shipping speeds, code validation, order totals preview, and payment card inputs.
- **Coupons:** Interactive promo code entry during checkout. Auto-validates active coupon codes against database values, applying fixed or percentage discounts with itemized tax and shipping fees.
- **Mock Payment Intent:** Uses a custom `/api/payment/create-intent` endpoint representing a production Stripe handshake returning a client secret to proceed safely.

2.5 Order Management & Tracking

- **Timeline Tracking:** Visual stepper track illustrating order processing history (`Placed` → `Confirmed` → `Packed` → `Shipped` → `Out for Delivery` → `Delivered`).
- **Invoice Overview:** Order history details, itemized totals, discount summary, and delivery statuses.

2.6 Interactive Customer Reviews

- **Verified Purchase:** Automatically flags reviews left by users who actually purchased the product.
- **Upvote System:** Customers can vote reviews as "Helpful" which dynamically counts and highlights the helpful tally.

3. Administrative Control Ledger

Inside [AdminDashboard.jsx](#): - **Live Sales & Growth Charts**: Chart.js representations of monthly revenue, category breakdown performance, active user growth trend, and low stock warnings. - **Interactive Tables**: - **Products**: Complete CRUD controls for titles, prices, descriptions, and stock quantities (with alert thresholds). - **Orders**: A live fulfillment selector allowing administrative status overriding. - **Customers**: Toggles for blocking or unblocking customer accounts instantly. - **Coupons**: Custom coupon manager supporting creation, editing, status toggles, and deletion.

4. Verification & Clean Builds

- **Vite Build**: Compiled successfully in `2.81s`, outputting standard `dist` production chunks: `bash`
`dist/index.html 0.47 kB dist/assets/index-FhdFM_lc.css 58.46 kB`
`dist/assets/index-t8kN6ACC.js 660.89 kB`
- **Code Health**: All imports, Route paths, Context states (Auth, Cart, Toast), and Tailwind configurations are verified and clean.